MIL-I-22479(Wep)
30 March 1960
SUPERSEDING
NAVORD OS 7193A
3 April 1958

MILITARY SPECIFICATION

IGNITER, ELECTRIC

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope. - This specification covers electric igniters suitable for the initiation of certain pyrotechnic compositions.

1.2 Classification. - The igniters shall be of four types which differ only in the lead length.

Type I - Igniter, Electric (LD 299834) - 5.00 inch leads
Type II - Igniter, Electric (LD 299879) - 7.00 inch leads
Type III - Igniter, Electric (LD 479280) - 5.312 inch leads
Type IV - Igniter, Electric (LD 479281) - 3.500 inch leads

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

PPP-B-676	Boxes, Set-up, Paperboard
PPP-B-621	Boxes, Wood, Nailed and Lock-corner
PPP-B-636	Boxes, Fiber

FSC 1370

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-202	Test Methods for Electronic and Electric Component Parts
MIL-P-116	Preservation, Methods of
MIL-B-131	Barrier Material; Water Vapor-proof, Flexible

DRAWINGS

BUREAU OF ORDNANCE

LD 299834	Igniter, Electric
LD 299879	Igniter, Electric
LD 479280	Igniter, Electric
LD 479281	Igniter, Electric

And all documents listed in the above Lists of Drawings.

PUBLICATIONS

BUREAU OF ORDNANCE

OP 400	General Instructions for the Design, Manufacture and Inspection of Naval Ordnance Equipment

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on the date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-78	Transportation, Interstate Commerce Commission, Explosives and Other Dangerous Articles

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(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations. Applications for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should request the latest edition and supplements thereto.)

3. REQUIREMENTS

3.1 General

3.1.1 The igniters covered by this document shall be manufactured in accordance with LD 299834, LD 299879, LD 479280 and LD 479281 as applicable and all documents listed thereon.

3.1.2 The requirements and procedures established in Publication OP 400 shall be adhered to wherever applicable.

3.1.3 Preproduction sample. - Unless otherwise specified in the contract or order, a preproduction sample shall be submitted prior to entering into full production. The units shall be manufactured by the same procedures and processes and at the same location proposed for the production lot. The provisions of 4.2 are applicable for the delivery and acceptance of the sample.

3.2 Circuit resistance. - Circuit resistance limits prescribed by the drawing shall be measured in a manner that limits the bridge wire current of igniters accepted for service use to a maximum value of 50 milliamperes. The provisions of 4.5.1 shall be applicable for testing.

3.3 Non-functioning current. - The igniter shall not function when subjected to direct current of 270 milliamperes at any temperature between the limits minus 65 and 165 degrees Fahrenheit. Acceptance shall be governed by 4.5.2.

3.4 Functioning. - The igniter shall function to produce a visible flash within 100 milliseconds after the leads are subjected to the discharge of a 4000 ± 1.0 percent microfarad capacitor charged to 1.9 ± .05 volts. Satisfactory functioning shall be obtainable at any temperature between the limits of minus 65 and 165 degrees Fahrenheit. Except for preliminary lots and specially directed programs conformance with 4.5.3 will be accepted as full compliance with this requirement.

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3.5 Workmanship. - The match head face plates shall be cleanly cut and free from fins. The assembly shall be clean and free from corrosion, fluxing agents and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent which is employed. The coating of the match head shall be smooth, unbroken and free from blush. The leads shall be flat without bends, kinks, or nicks.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling. - Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government inspector in the determination of the acceptability of products submitted by a supplier for Government inspection shall be in accordance with the provisions of Standard MIL-STD-105.

4.1.1 Lot. - As applied to the Government inspection of units of product, the term "lot" shall mean "inspection lot", i.e., a collection of units of the product used as a basis for Government inspection. Unless otherwise specified, the number of units of product in inspection lots shall be as determined by the Government inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment or other purpose.

4.2 Pre-production sample. - At the start of pilot production (or prior to entering into full production, if there is no pilot production) a preproduction sample of 200 igniters shall be delivered by the contractor to the Government inspector. These igniters shall be manufactured by the same procedures and processes and at the same location proposed by the contractor for full production, using the same tooling, fixtures, and equipment as for full production. The Government inspector shall perform, witness, or participate in such inspections and tests on the samples as are required for full production. Upon conditional acceptance by the inspector, the pre-production sample shall be forwarded together with a report of inspection to an activity designated by the procuring agency for final inspection and acceptance. If the sample complies with all requirements of the contract, specifications and drawings, the contractor will be notified and will be authorized by the procuring activity to proceed with the submission of the product for acceptance. The preproduction sample shall be prepared for delivery in accordance with the applicable section of this document. Accepted samples shall become the property of the Government and shall be included in the number called for in the contract schedule.

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4.2.1 Pre-production sample for a subsequent contract. - The necessity for a pre-production sample will be determined by the Government when production under a new contract by the same contractor at the same location follows the manufacture of satisfactory material as covered by this document.

4.3 Production lot. - For procurement purposes each lot shall consist of 2000 igniters plus those required for all test purposes. Unless otherwise stated in the contract, order or requisition, all units for test purposes shall be at the expense of the contractor. The production lot should not be confused with the lot for inspection purposes as defined in Standard MIL-STD-105.

4.4 Inspection. - The general inspection procedures shall be in accordance with the applicable provisions of Publication OP 400. Government inspection shall conform with the General Procedures for Government Inspection and contractor's inspection shall be in accordance with Inspection by Contractor as contained in Publication OP 400.

4.5 Tests.

4.5.1 Circuit resistance test. - The circuit resistance shall be measured for conformity with the requirements of the drawings in accordance with Method 303 of Standard MIL-STD-202. The measuring current shall be less than 50 milliamperes.

4.5.2 Nonfunctioning current test. - Sampling and disposition of the lot shall be in accordance with Standard MIL-STD-105 and Appendix, Inspection Level L7, AQL 1.0 percent defective. Subject the sample to conditioning for 24 hours at a temperature of 160 to 165 degrees Fahrenheit. After conditioning, subject each igniter circuit to a current of 260 to 270 milliamperes for at least 30 seconds while at the conditioning temperature. An igniter which burns or visibly decomposes as a result of conditioning or current passage shall be considered a defective. The test is destructive and the sample shall not be included in accepted lots.

4.5.3 Functioning test. - Sampling and disposition of the lot shall be in accordance with Standard MIL-STD-105, Inspection Level II, AQL 0.25 percent defective. The provisions of the standard for tightened and reduced inspection are applicable in lot disposition. Condition the sample for a period of 24 hours at a temperature of minus 55 to minus 65 degrees Fahrenheit. The test chamber shall be equipped with a vision port and external electrical connections for the igniters. The sample shall be introduced into the cold chamber in a manner which avoids contamination

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with condensation and frost. Initiate the igniters by means of Capacitor Test Firing Circuit Drawing 1282014 which is a part of LD 299834, LD 299879, LD 479280 and LD 479281. An igniter which does not ignite with a visible flash shall be considered a defective. Where delay functioning observable without instrumentation is encountered, the lot shall be resampled. The sample shall be submitted to a Government laboratory for instrumental testing of functioning within the 100 millisecond maximum functioning time limit.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A. (Not applicable)

5.1.2 Level B. (Not applicable)

5.1.3 Level C.

5.1.3.1 Preservative application. (Not applicable)

5.1.3.2 Packaging. - One hundred (100) electric igniters shall be packaged Method IA-15 in accordance with Military Specification MIL-P-116.

5.1.3.2.1 One hundred (100) electric igniters shall be packaged in a paperboard box conforming to Federal Specification PPP-B-676, Type IV. A non-hygroscopic cushioning material shall be employed to prevent the kinking, bending, marring or deforming of the metallic leads; to prevent direct contact of the inflammable portion of the igniter with each other or any other frictional material; and to insure a tight package. Each paperboard box shall then be enclosed in a heat sealed bag fabricated from barrier material conforming to Specification MIL-B-131.

5.1.3.3 Intermediate package. - Five boxes (500 igniters) shall be intermediately packaged in a corrugated fiberboard box in accordance with Specification LLL-B-631.

5.2 Packing.

5.2.1 Level A. (Not applicable)

5.2.2 Level B. (Not applicable)

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5.2.3 Level C. - Five hundred (500) igniters packaged as described in 5.1.3.2 and 5.1.3.3 shall be overpacked in a nailed wood box in accordance with Specification PPP-B-621, Class 1.

5.3 Marking.

5.3.1 In addition to any special marking required by the contract or order, all marking shall be in accordance with Military Standard MIL-STD-129 and the Code of Federal Regulations 49 CFR 71-78.

6. NOTES

6.1 Ordering data. - Procurement documents should specify:

(a) Title, symbol and date of this specification.
(b) Type of igniter under procurement.
(c) Government laboratory or agency where pre-production samples should be forwarded (see 4.2).

6.2 General safety precautions. - The loading, assembly and handling of explosives, ammunition, ammunition components, fuzes, detonators, primers, tracers, boosters, pyrotechnics, etc., and subassemblies thereof, involve hazardous operations and therefore require suitable explosives safety precautions. Use of this specification shall not be construed as to relieve the contractor or manufacturer of responsibility for the safety of his operations. Listed below are the minimum provisions which a contractor or manufacturer (loader) of the above items should observe in order to fulfill his responsibility for safety. At Bureau of Ordnance, Navy Department, and other government plants, these provisions are mandatory. Such other warnings and precautions, pertinent to the operational effectiveness or safety during use or loading of the specified item, are included in the detail provisions of the specification.

6.2.1 All loading operations should be conducted in a neat and orderly manner.

6.2.2 Safe equipment and methods should be utilized for transporting and handling explosives and loaded parts. Where required, remote control barricaded handling equipment shall be used for explosives operations, such as mixing, pouring, weighing, charging, sifting, drying, pressing, casting, crimping, etc.

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6.2.3 Personnel handling detonators, primers, delay elements, lead-ins, boosters, and related parts which effect functioning, should in so far as practicable, avoid using bare fingers or improper equipment in order to prevent damage, corrosion, or deterioration from perspiration or other contaminating deposits.

6.2.4 The exposure of explosive materials and related parts should be so controlled as to minimize the absorption of moisture from the atmosphere or other sources during loading and handling operations.

6.2.5 All explosives and completely or partially loaded items should be stored in suitable storage magazines located in accordance with the American Table of Distances (ATD) or other applicable safety standards; and, while in process, in safety lockers and chests if in loading rooms, or in adequate ready or service magazines located in accordance with intra-plant distances when outside of loading rooms. For Navy explosives loading plants, the provisions of the Armed Services Explosives Safety Board covering quantity-distance relations for explosives, shall apply.

6.2.6 Proper care should be exercised at all times to protect personnel from accidents, fires or explosions, and to limit damage to equipment and loading areas. In this connection, the following precautionary measures should be observed:

6.2.6.1 Employ properly proportioned and properly located protective barricades, screens or shields at all required points.

6.2.6.2 Keep only minimum limited quantities of explosives and completed or partially loaded parts present at each stage of operation.

6.2.6.3 Keep explosives and explosive parts in approved covered receptacles with covers in place when material is not being taken out of or put into the receptacles. Where necessary, receptacles should be conductive to ground electrostatic charges.

6.2.6.4 Protect operations from electrostatic charges by effectively grounding all machinery, equipment, and fixtures; and, where necessary, employ suitable grounded conductive coverings for floors, work benches and tables, and workers' conductive shoes. Workers' clothing of a type to minimize the accumulation of static charges should be employed. Fabrics such as silk and nylon, which promote static generation should be avoided. Additional grounding devices such as grounded bracelets for workers should be employed where operations are conducted with items which are unusually sensitive to initiation by static electricity. Such items

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include initiating explosives, tracer mixtures, and low-energy type electric primers, detonators and squibs. The latter types of items should have the free ends of lead wires bared and twisted together, and be packed in relatively small groups wrapped in bare non-insulated aluminum foil or other uncoated metal foil. During assembly and processing operations such sensitive electric items should be short circuited by clips or other devices until installed with safety shunt in the final device. Additional precautions for these items should include mechanical shielding to contain or deflect fragments and blast, also electrical shielding of these items from induced electric currents generated by sources such as lightning, static, radiations from communications apparatus, radar, or high frequency heating apparatus, etc. Where necessary for safety, humidity (See 6. 2. 4) of work rooms should be increased, as required to lessen electrostatic effects without excessive moisture absorption.

6. 2. 6. 5 Protect all explosive operations from effects of electric current originating from equipment such as soldering irons, heaters, switches, wiring, motors, lights, test instruments, etc., by suitable insulation, grounding, separation or shielding. Such electric sources may initiate explosives by heat, sparks, arcs, or due to completing an electric circuit through an electric primer, detonator, or squib. Circuits may be inadvertently completed, for example, from a defective electric soldering iron through a grounded contact. All electric type primers, detonators or squibs provided with wire leads should have the free ends of the wires bared and twisted together to short circuit each unit, except when in process of assembly into a finished item. Where practicable, removable short circuiting clips, or other devices should be employed during manufacturing operations involving electric primers, detonators or squibs.

6. 2. 6. 6 Enforce, where necessary, the wearing of suitable safety footwear, gloves, goggles, respirators, and impregnated garments to protect personnel against burns, poisoning and associated industrial hazards.

6. 2. 6. 7 Allow no fires or exposed electrical or other sparking equipment, and little or no flammable material to be present in loading, handling and storage spaces. Enforce proper "Match" and "No Smoking" rules where necessary.

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6. 2. 6. 8 Enforce good housekeeping and maintain effective policing, inspection and supervisory methods throughout the loading area and surroundings. Employ effective cleaning methods periodically to minimize the accumulation of explosives or explosive dust and other contamination upon, and assure its removal from floors, walls, ceilings, ledges, tables, benches, piping, and equipment or the items loaded; also, clean up any spilled material immediately.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any invention that may in any way be related thereto.

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